

Git Hands-On-1

Step 1: Setup my machine with Git Configuration

```
$ git --version
```

Output:

```
git version 2.45.1.windows.1
```

Next, I set my user name and email so that my commits would be identified correctly:

```
$ git config --global user.name "Student Name"
```

```
$ git config --global user.email "student.name@example.com"
```

To verify that the configuration was saved correctly, I ran:

```
$ git config --list
```

Output:

```
user.name=Student Name
user.email=student.name@example.com
core.autocrlf=true
core.fscache=true
core.symlinks=false
```

This confirmed that my user name and email were correctly set at the global (user) level.

Step 2: Integrate notepad++.exe with Git and make it the default editor

```
$ notepad++
```

Output:

```
bash: notepad++: command not found
```

After closing and reopening Git Bash, I ran notepad++ again:

```
$ notepad++
```

Output:

```
$ echo 'alias notepad++="/c/Program\ Files/Notepad++/notepad++.exe"' >> ~/.bashrc
$ source ~/.bashrc
```

Then I configured notepad++ as Git's default editor:

```
$ git config --global core.editor "notepad++ -multiInst -notabbar -nosession -noPlugin"
```

To verify notepad++ was set as the default editor, I ran:

```
$ git config --global -e
```

Output:

```
[user]
  name = Student Name
  email = student.name@example.com
[core]
  editor = notepad++ -multiInst -notabbar -nosession -noPlugin
```

Step 3: Add a file to the source code repository

```
$ git init GitDemo
```

Output:

```
Initialized empty Git repository in C:/Users/Student/GitDemo/.git/
$ cd GitDemo
```

To verify the repository was initialized and to view the hidden .git folder, I executed:

```
$ ls -la
```

Output:

```
total 4
drwxr-xr-x 1 Student 197121 0 Jul 25 10:02 .
drwxr-xr-x 1 Student 197121 0 Jul 25 10:02 ..
drwxr-xr-x 1 Student 197121 0 Jul 25 10:02 .git
```

Next, I created a file named welcome.txt and added some content to it:

```
$ echo "Welcome to Git Hands-on Lab" > welcome.txt
```

To verify the file was created, I ran:

```
$ ls -la
```

Output:

```
total 8
drwxr-xr-x 1 Student 197121 0 Jul 25 10:05 .
drwxr-xr-x 1 Student 197121 0 Jul 25 10:02 ..
drwxr-xr-x 1 Student 197121 0 Jul 25 10:02 .git
-rw-r--r-- 1 Student 197121 29 Jul 25 10:05 welcome.txt
```

To verify the content of the file, I ran:

```
$ cat welcome.txt
```

Output:

```
Welcome to Git Hands-on Lab
```

```
$ git status
```

Output:

```
On branch master

No commits yet

Untracked files:
```

```
(use "git add <file>..." to include in what will be committed)
welcome.txt
```

```
nothing added to commit but untracked files present (use "git add" to track)
```

This showed that welcome.txt was present in the working directory but was not yet tracked by Git.

To make welcome.txt tracked by the Git repository, I added it to the staging area:

```
$ git add welcome.txt
```

To add a multi-line commit message, I opened the default editor (notepad++) by executing:

```
$ git commit
```

Notepad++ opened, and I typed the following commit message:

```
Added welcome.txt file
```

```
This is the first commit for the GitDemo project.
It adds a welcome.txt file with a greeting message.
```

After saving and closing the editor, the commit was created and Git Bash displayed:

Output:

```
[master (root-commit) 3f2b9a1] Added welcome.txt file
1 file changed, 1 insertion(+)
create mode 100644 welcome.txt
```

To check that the local repository and working directory were now in sync, I ran:

```
$ git status
```

Output:

```
On branch master
nothing to commit, working tree clean
```

This confirmed that welcome.txt was successfully added and committed to the local Git repository.

I then signed up on GitLab and created a remote repository named “GitDemo”, and linked it to my local repository:

```
$ git remote add origin https://gitlab.com/studentname/GitDemo.git
```

To pull from the remote repository (to sync any changes made on GitLab, e.g. a README):

```
$ git pull origin master
```

Output:

```
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0)
Unpacking objects: 100% (3/3), 682 bytes | 34.00 KiB/s, done.
From https://gitlab.com/studentname/GitDemo
 * branch          master      -> FETCH_HEAD
 * [new branch]    master      -> origin/master
Merge made by the 'ort' strategy.
 README.md | 1 +
```

```
1 file changed, 1 insertion(+)  
create mode 100644 README.md
```

Finally, I pushed my local commits (including welcome.txt) up to the remote GitLab repository:

```
$ git push origin master
```

Output:

```
Enumerating objects: 4, done.  
Counting objects: 100% (4/4), done.  
Delta compression using up to 8 threads  
Compressing objects: 100% (2/2), done.  
Writing objects: 100% (3/3), 331 bytes | 331.00 KiB/s, done.  
Total 3 (delta 0), reused 0 (delta 0)  
To https://gitlab.com/studentname/GitDemo.git  
9c1a2ef..3f2b9a1 master -> master
```